

Woojin Bae

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EDUCATION

Seoul National University (SNU), Seoul, South Korea M.S. Chemical and Biological Engineering (CBE) (PI: Prof. Yousung Jung)	Aug. 2028 (expected)
Seoul National University, Seoul, South Korea B.S., CBE, Cum Laude	Aug. 2026
North Carolina State University (NCSU), Raleigh, NC Exchange student in Chemical and Biomolecular Engineering	May 2025
Daejeon Science High School for the Gifted, Daejeon, South Korea	Feb. 2020

PUBLICATIONS & PRESENTATIONS (†: equally contributed, *: corresponding author)

J. Hong[†], H. Jang[†], **W. Bae[†]**, Jaewon Bae*, Chanyoung Park.*, Retro-Forge: A Multi-Step Pairwise Retrosynthesis Framework for Solid-State Materials Synthesis. *ICML 2026 Workshop AI4Science*.

Jinge Xu, Junbin Li, Pragyan Jha, Nikolai Mukhin, **Woojin Bae**, Christopher H.J. Moran, Richard B. Canty, Milad Abolhasani*, Self-Driving Chemistry for Route-Encoded Colloidal Nanoscience. *Submitted to Nature Synthesis*

W. Bae[†], S.Jung[†], S. Ji[†], Y. Jeon* & J. Park*, SEI Structures Dependent Li-Morphology Characterized by Cryo-TEM. [Poster Presentation]

- 2025 KICChE Fall meeting and International Symposium, Oct. 15-18, 2025.
- The 2024 Annual Fall Conference of Korean Society of Microscopy, Nov. 25-26, 2024.

RESEARCH EXPERIENCE

NanoForge AI - Researcher Mar. 2026 – Aug. 2026

Topic: Solid state reaction process prediction leveraging Large Language Model.

- Trained an LLM model to predict precursor selection as a stepwise solid-state reaction planning task.
- Leveraged chemical prior and *Retrieval Augmented Generation* (RAG) to predict optimal calcination and sintering temperatures.

Multi-dimensional Materials Chemistry Lab, SNU Mar. 2024 – Mar. 2026
Undergraduate Researcher - (PI: Prof. Jungwon Park) Seoul, Korea

Topic: SEI Structures Dependent Li-Morphology Characterized by cryo-TEM

- Team Leader; Developed a DBSCAN-based protocol for quantitative image analysis protocol for cryogenic TEM characterization of SEI structures.
- Observed correlation between SEI morphology, crystallinity, and composition; proposed a formation mechanism of Li-deposition. (Co-Advised by Prof. Won Bo Lee)

Topic: Computational Modeling of AI-Driven Synthesized Perovskite Nanocrystals

- Awarded a \$2,000 research grant by SNU College for a project across SNU and NCSU.
- Initiated and led collaborative project across SNU and NCSU to characterize AI-driven synthesized Mn-doped CsPb(Cl/Br)₃ QDs, by TEM analysis and DFT calculation.

Self-Driving Fluidic Lab, NCSU Jan. 2025 – May 2025
Undergraduate Researcher - (PI: Prof. Milad Abolhasani) Raleigh, NC

Topic: Autonomous Mn-doped Perovskite Nanocrystals Synthesis by a Self-Driving Lab

- Achieved Photoluminescence Quantum Yield improvement of Mn-doped CsPb(Cl/Br)₃ perovskite through genetic algorithm-based synthesis optimization in autonomous lab.
- Presented a poster at Spring Undergraduate Research & Creativity Symposium.

Summer Undergraduate Initiative (SURI), ASU

Undergraduate Researcher - (PI: *Prof. Sandhya Susarla*)

May 2025 – Jul. 2025

Tempe, AZ

Topic: Synthesis of Perovskite QDs for Phonon-Polariton Interaction with hBN

- Selected for the SURI fellowship at Arizona State University (\$5,000 stipend).
- Synthesized perovskite nanocrystals with controlled morphologies to investigate phonon-polariton coupling with hexagonal boron nitride (hBN).
- Presented a poster at Summer Research Initiative Final Research Showcase.

HONORS & AWARDS

Honorable Mention | Korea Process Simulation Olympiad 2025, KICChE, Korea *Sep. 2025*

- National finalist with process design for high-purity methanol (PI: *Prof. Jong Min Lee*)

Excellence Research Award | 2024 Student-Directed Research Symposium *Jan. 2025*

KorUS Educational Exchange Initiative for Youth in STEM | MOTIE, Korea *Nov. 2024*

- Awarded national \$9,000 scholarship as a STEM-specialized exchange student.

Excellence Award | Undergraduate Research Program for Advanced Equipment *Jan. 2024*

Team Section Winner | We Change SNU Contest, SNU *Jul. 2022*

- A spatial-based messenger allowing messages to be accessed only at SNUfestival locations designated by the sender.

Education and Research Foundation Scholarship | College of Engineering, SNU *Spring 2021*

Future Chemical Talent Award | Korean Union of Chemical Science & Technology *Feb. 2020*

- Awarded as one of the top 6 science high school students in Korea.

LEADERSHIP & ACTIVITIES

Undergraduate Member | Korean-American Scientists and Engineers Association *Jan. 2025*

- Delivered a research presentation at Korean-American undergraduate conference.

Tokyo Forum Youth Session | The University of Tokyo, Japan *Fall 2024*

- Representative of Korea, in a session on Aging Society and Low Birth Rate.

Young Engineers Honor Society | National Academy of Engineering of Korea *Nov. 2024*

- Korean engineering honor society under National Academy of Engineering of Korea.

Entrepreneurship Bootcamp | Hong Kong University of Science and Technology *Jul. 2024*

- Led a global team in a mock-up competition and won third prize for developing a worldwide healthcare platform among international participants.

SNU-UMN Culture Exchange Program | University of Minnesota Twin Cities, MN *Mar. 2024*

Winter Data Science Bootcamp: **Computing** | Graduate School of Data Science, SNU *Feb. 2024*

SNU in EU Union | Brussels School of Governance, Belgium *Jul. 2022*

- Study Abroad Program, delivered a presentation titled "The European External Action Service".

Republic of Korea Air Force | 11th Fighter Wing, Daegu, Korea *Aug. 2022 – May 2024*

Marketing team manager | SNUfestival, Student Council, SNU *Jun. 2021 – May 2022*

- Promoted the festival through the online website and secured \$2,200 sponsorship from a shared scooter company during the first in-person festival after the COVID-19 pandemic.

Vice president | Chemical and Biological Engineering Interim Leadership, SNU *Winter 2020*

- Developed an online program for incoming freshmen during the COVID-19 pandemic.

TEACHING & MENTORING EXPERIENCES

Peer Tutor | Basic Computing: Exploring Data for the first time, SNU *Spring 2026*

SNU Corps in Indonesia | Universitas Mataram, Lombok, Indonesia *Feb. 2026*

Korean Language Peer Mentor | Language Education Institute, SNU *Fall 2024*

Peer Tutor | Basic Computing: First Adventures in Computing, SNU *Fall 2024, 2025*

Lecturer, TEM Seminar | Multi-dimensional Materials Chemistry Lab, SNU *May 2024*

Student Mentor, SNU Mentoring | SNU Social Responsibility *Apr. 2023 – Aug. 2025*

TECHNICAL SKILLS & LANGUAGES

Instruments: TEM, SEM, XRD, STEM/EDS, UV-Vis, OM, PL, ICP-MS

Programming Language: Python, MATLAB, C

Software: Gatan DigitalMicrograph, VASP, APS, Adobe, Fusion 360

Languages: Korean (Native), English (C1, TOEFL 103)